BIG DATA ANALYTICS LAB

(Course Code: 23UPCSC1L05)

A laboratory record submitted to Periyar University, Salem

In partial fulfillment of the requirements for the degree of

MASTER OF COMPUTER APPLICATION

By

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(NAAC `A++` Grade with CGPA 3.61) – NIRF RANK 56 – ARIIA RANK 10

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SALEM – 636 011.

(APRIL - 2024)

CERTIFICATE

This is to certify that the Programming Laboratory entitled "**BIG DATA ANALYTICS LAB (23UPCSC1L05)"** is a bonafide record work done by Mr./Ms._____Register No: _____ as partial fulfillment of the requirements for the degree of Master of Computer Application, in the Department of Computer Science, Periyar University, Salem, during the Academic Year 2024-2026.

Staff In-charge Head of the Department

Submitted for the practical examination held on……..……

Internal Examiner External Examiner

CONTENTS

S.NO	DATE	TITLE OF THE PROGRAM	PAGE NO	SIGNATURE
1.		Implement File System Shell Commandsfor HDFS in Hadoop Environment		
2.		Write a Mapreduce program using single reduce function for finding Maximum andMinimum Number		
3.		Write a Mapreduce program using multiplereduce function for Word Count in an givenText document		
4.		Implement the following using Pig LatinInput and Output Operations Relational Operations		
5.		Implement the following using Pig LatinUser Defined Functions Advanced Relational Operations		
6.		Write a Word Count program using PigLatin Script		
7.		Write a program to find a maximumtemperature using Pig Latin Script		
8.		Implement the following using Hive commands Handling the Database Creatingand Manipulating table		
9.		Implement Simple Queries for databaseusing MongoDB		
10.		Implement Simple Queries for collectionsusing MongoDB		

SOURCE CODE:

rr@ubuntu:~$ start-all.sh
This script is Deprecated. Instead use start-dfs.sh and start-yarn.sh
24/04/03 04:09:01 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting namenodes on [localhost]
localhost: starting namenode, logging to /usr/local/hadoop/logs/hadoop-rr-namenode-ubuntu.out
localhost: starting datanode, logging to /usr/local/hadoop/logs/hadoop-rr-datanode-ubuntu.out
Starting secondary namenodes [0.0.0.0]
0.0.0.0: starting secondarynamenode, logging to /usr/local/hadoop/logs/hadoop-rr-secondarynamenode-ubuntu.out
24/04/03 04:09:36 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
starting yarn daemons
starting resourcemanager, logging to /usr/local/hadoop/logs/yarn-rr-resourcemanager-ubuntu.out
localhost: starting nodemanager, logging to /usr/local/hadoop/logs/yarn-rr-nodemanager-ubuntu.out

rr@ubuntu:~$ jps
6581 NameNode
7081 ResourceManager
6716 DataNode
7533 Jps
7230 NodeManager
6943 SecondaryNameNode

rr@ubuntu:~$ hdfs dfs -ls /home
24/04/03 04:10:42 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
drwxr-xr-x - rr supergroup 0 2024-03-27 05:10 /home/rr

rr@ubuntu:~$ hdfs dfsadmin -safemode leave
24/04/03 04:12:30 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Safe mode is OFF

rr@ubuntu:~$ hdfs dfs -rmr /home/rr
rmr: DEPRECATED: Please use 'rm -r' instead.
24/04/03 04:12:40 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
24/04/03 04:12:42 INFO fs.TrashPolicyDefault: Namenode trash configuration: Deletion interval = 0 minutes, Emptier interval = 0 minutes.
Deleted /home/rr

rr@ubuntu:~$ hdfs dfs -ls /home
24/04/03 04:12:51 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

rr@ubuntu:~$ hdfs dfs -mkdir /home/hari/
24/04/03 04:13:26 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

rr@ubuntu:~$ hdfs dfs -ls /home
24/04/03 04:13:35 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
drwxr-xr-x - rr supergroup 0 2024-04-03 04:13 /home/hari

rr@ubuntu:~$ hdfs dfs -put data.txt /home/hari
24/04/03 04:16:24 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicablerr@ubuntu:~$ hdfs dfs -ls /home/hari
24/04/03 04:18:09 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r-- 1 rr supergroup 244 2024-04-03 04:16 /home/hari/data.txt

rr@ubuntu:~$ hdfs dfs -cat /home/hari/data.txt
24/04/03 04:18:28 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Whether used in health care, government, finance, or some other industry, big data analytics is behind some of the most significant industry advancements in the world today. Read on to learn more about big data analytics and its many benefits.

rr@ubuntu:~$ hdfs dfs -get /home/hari/data.txt /home/rr/
24/04/03 04:27:20 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
24/04/03 04:27:23 WARN hdfs.DFSClient: DFSInputStream has been closed already

rr@ubuntu:~$ hdfs dfs -du /home/hari/
24/04/03 04:27:40 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
244 /home/hari/data.txt

rr@ubuntu:~$ stop-all.sh
This script is Deprecated. Instead use stop-dfs.sh and stop-yarn.sh
24/04/03 04:28:02 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Stopping namenodes on [localhost]
localhost: stopping namenode
localhost: stopping datanode
Stopping secondary namenodes [0.0.0.0]
0.0.0.0: stopping secondarynamenode

```
24/04/03 04:28:45 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
stopping yarn daemons
stopping resourcemanager
localhost: stopping nodemanager
no proxyserver to stop
```

SOURCE CODE:

rr@ubuntu:~$ start-all.sh

```
This script is Deprecated. Instead use start-dfs.sh and start-yarn.sh
24/04/03 04:40:06 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
Starting namenodes on [localhost]
localhost: starting namenode, logging to /usr/local/hadoop/logs/hadoop-rr-namenode-
ubuntu.out
localhost: starting datanode, logging to /usr/local/hadoop/logs/hadoop-rr-datanode-ubuntu.out
Starting secondary namenodes [0.0.0.0]
0.0.0.0: starting secondarynamenode, logging to /usr/local/hadoop/logs/hadoop-rr-
secondarynamenode-ubuntu.out
24/04/03 04:40:53 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
starting yarn daemons
starting resourcemanager, logging to /usr/local/hadoop/logs/yarn-rr-resourcemanager-
ubuntu.out
localhost: starting nodemanager, logging to /usr/local/hadoop/logs/yarn-rr-nodemanager-
ubuntu.out
```

rr@ubuntu:~$ gedit marketpricedata.txt

```
2011 1599
2011 909
2011 859
2012 1650
2012 980
2012 2584
2013 1195
2013 1856
2013 1958
2014 3568
2014 2785
2014 2854
2015 4561
2015 2398
```

rr@ubuntu:~$ gedit minMaxMapper.py

```
import  sys
for line in sys.stdin:
   val=line.strip()
   year=val[0:4]
   temp=val[5:9]
   if temp!='9999':
      print('%s\t%s' %(year,temp))
```

rr@ubuntu:~$ gedit minMaxReducer.py

```
from operator import itemgetter
import sys
```

```
(last_key,max_val,min_val)=(None,0,0)

for line in sys.stdin:
    (key,val)=line.strip().split("\t")

    if last_key and last_key!=key:
        print('%s\t%s\t%s' %(last_key,max_val,min_val))
        (last_key,max_val,min_val)=(key,float(val),float(val))
    else:
        (last_key,max_val,min_val)=(key,max(max_val,float(val)),min(min_val,float(val)))
        if min_val==0.0:
            min_val=max_val

if last_key:
    print('%s\t%s\t%s' %(last_key,max_val,min_val))
```

rr@ubuntu:~$ hdfs dfsadmin -safemode leave
24/04/03 04:49:11 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Safe mode is OFF

rr@ubuntu:~$ hdfs dfs -put marketpricedata.txt /home/hari/
24/04/03 04:50:02 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

rr@ubuntu:~$ hdfs dfs -put minMaxMapper.py /home/hari/
24/04/03 04:50:46 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

rr@ubuntu:~$ hdfs dfs -put minMaxReducer.py /home/hari/
24/04/03 04:51:09 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

rr@ubuntu:~$ hdfs dfs -ls /home/hari/
24/04/03 04:51:24 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 4 items
```
-rw-r--r--   1 rr supergroup        137 2024-04-03 04:50 /home/hari/marketpricedata.txt
-rw-r--r--   1 rr supergroup        144 2024-04-03 04:50 /home/hari/minMaxMapper.py
-rw-r--r--   1 rr supergroup        525 2024-04-03 04:51 /home/hari/minMaxReducer.py
```

rr@ubuntu:~$ hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-*.jar -file minMaxMapper.py -mapper "python minMaxMapper.py" -file minMaxReducer.py -reducer "python minMaxReducer.py" -input /home/hari/marketpricedata.txt -output /home/hari/result
24/04/03 04:58:11 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
24/04/03 04:58:12 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

```
packageJobJar: [minMaxMapper.py, minMaxReducer.py, /tmp/hadoop-
unjar3959138942582892722/] [] /tmp/streamjob1459118211949816704.jar tmpDir=null
24/04/03 04:58:15 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
24/04/03 04:58:15 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
24/04/03 04:58:17 INFO mapred.FileInputFormat: Total input paths to process : 1
24/04/03 04:58:17 INFO mapreduce.JobSubmitter: number of splits:2
24/04/03 04:58:18 INFO mapreduce.JobSubmitter: Submitting tokens for job:
job_1712148067987_0002
24/04/03 04:58:18 INFO impl.YarnClientImpl: Submitted application
application_1712148067987_0002
24/04/03 04:58:18 INFO mapreduce.Job: The url to track the job:
http://localhost:8088/proxy/application_1712148067987_0002/
24/04/03 04:58:18 INFO mapreduce.Job: Running job: job_1712148067987_0002
24/04/03 04:58:32 INFO mapreduce.Job: Job job_1712148067987_0002 running in uber
mode : false
24/04/03 04:58:32 INFO mapreduce.Job:  map 0% reduce 0%
24/04/03 04:59:00 INFO mapreduce.Job:  map 50% reduce 0%
24/04/03 04:59:02 INFO mapreduce.Job:  map 100% reduce 0%
24/04/03 04:59:17 INFO mapreduce.Job:  map 100% reduce 100%
24/04/03 04:59:18 INFO mapreduce.Job: Job job_1712148067987_0002 completed
successfully
24/04/03 04:59:19 INFO mapreduce.Job: Counters: 49
        File System Counters
                FILE: Number of bytes read=171
                FILE: Number of bytes written=356210
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=416
                HDFS: Number of bytes written=93
                HDFS: Number of read operations=9
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
        Job Counters
                Launched map tasks=2
                Launched reduce tasks=1
                Data-local map tasks=2
                Total time spent by all maps in occupied slots (ms)=54301
                Total time spent by all reduces in occupied slots (ms)=13092
                Total time spent by all map tasks (ms)=54301
                Total time spent by all reduce tasks (ms)=13092
                Total vcore-seconds taken by all map tasks=54301
                Total vcore-seconds taken by all reduce tasks=13092
                Total megabyte-seconds taken by all map tasks=55604224
                Total megabyte-seconds taken by all reduce tasks=13406208
        Map-Reduce Framework
                Map input records=14
                Map output records=14
                Map output bytes=137
                Map output materialized bytes=177
```

```
                Input split bytes=210
                Combine input records=0
                Combine output records=0
                Reduce input groups=5
                Reduce shuffle bytes=177
                Reduce input records=14
                Reduce output records=5
                Spilled Records=28
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=837
                CPU time spent (ms)=5440
                Physical memory (bytes) snapshot=651087872
                Virtual memory (bytes) snapshot=2370928640
                Total committed heap usage (bytes)=303374336
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=206
        File Output Format Counters
                Bytes Written=93
24/04/03 04:59:19 INFO streaming.StreamJob: Output directory: /home/hari/result
```

rr@ubuntu:~$ hdfs dfs -ls /home/hari/result

```
24/04/03 04:59:49 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r--   1 rr supergroup          0 2024-04-03 04:59 /home/hari/result/_SUCCESS
-rw-r--r--   1 rr supergroup         93 2024-04-03 04:59 /home/hari/result/part-00000
```

rr@ubuntu:~$ hdfs dfs -cat /home/hari/result/part-00000

```
24/04/03 04:59:59 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
2011    1599.0 859.0
2012    2584.0 980.0
2013    1958.0 1195.0
2014    3568.0 2785.0
2015    4561.0 2398.0
```

rr@ubuntu:~$ stop-all.sh

```
This script is Deprecated. Instead use stop-dfs.sh and stop-yarn.sh
24/04/03 05:00:13 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
Stopping namenodes on [localhost]
```

```
localhost: stopping namenode
localhost: stopping datanode
Stopping secondary namenodes [0.0.0.0]
0.0.0.0: stopping secondarynamenode
24/04/03 05:00:46 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
stopping yarn daemons
stopping resourcemanager
localhost: stopping nodemanager
localhost: nodemanager did not stop gracefully after 5 seconds: killing with kill -9
no proxyserver to stop
```

SOURCE CODE:

rr@ubuntu:~$ start-all.sh

```
This script is Deprecated. Instead use start-dfs.sh and start-yarn.sh
24/04/03 05:07:57 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
Starting namenodes on [localhost]
localhost: starting namenode, logging to /usr/local/hadoop/logs/hadoop-rr-namenode-
ubuntu.out
localhost: starting datanode, logging to /usr/local/hadoop/logs/hadoop-rr-datanode-ubuntu.out
Starting secondary namenodes [0.0.0.0]
0.0.0.0: starting secondarynamenode, logging to /usr/local/hadoop/logs/hadoop-rr-
secondarynamenode-ubuntu.out
24/04/03 05:08:37 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
starting yarn daemons
starting resourcemanager, logging to /usr/local/hadoop/logs/yarn-rr-resourcemanager-
ubuntu.out
localhost: starting nodemanager, logging to /usr/local/hadoop/logs/yarn-rr-nodemanager-
ubuntu.out
```

rr@ubuntu:~$ gedit bigdata.txt

Whether used in health care, government, finance, or some other industry, big data analytics is behind some of the most significant industry advancements in the world today. Read on to learn more about big data analytics and its many benefits.

rr@ubuntu:~$ gedit wordMapper.py

```
import sys
for line in sys.stdin:
    line=line.strip()
    words=line.split()
    for word in words:
        print('%s\t%s' %(word,'1'))
```

rr@ubuntu:~$ gedit wordReducer.py

```
import sys
wordcount={}
for line in sys.stdin:
   word,count=line.strip().split('\t')
   try:
      wordcount[word]=wordcount[word]+int(count)
   except:
      wordcount[word]=int(count)
for word in wordcount.keys():
     print('%s\t%s' %(word,wordcount[word]))
```

rr@ubuntu:~$ hdfs dfsadmin -safemode leave

```
24/04/03 05:11:00 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
Safe mode is OFF
```

rr@ubuntu:~$ hdfs dfs -put wordMapper.py /home/hari/
24/04/03 05:14:36 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

rr@ubuntu:~$ hdfs dfs -put wordReducer.py /home/hari/
24/04/03 05:15:09 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

rr@ubuntu:~$ hdfs dfs -put bigdata.txt /home/hari/
24/04/03 05:15:39 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

rr@ubuntu:~$ hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-*.jar -file wordMapper.py -mapper "python wordMapper.py" -file wordReducer.py -reducer "python wordReducer.py" -input /home/hari/bigdata.txt -output /home/hari/result
24/04/03 05:16:50 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
24/04/03 05:16:51 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
packageJobJar: [wordMapper.py, wordReducer.py, /tmp/hadoop-unjar4131138487846518480/] [] /tmp/streamjob3819747224130267099.jar tmpDir=null
24/04/03 05:16:54 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
24/04/03 05:16:54 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
24/04/03 05:16:56 INFO mapred.FileInputFormat: Total input paths to process : 1
24/04/03 05:16:56 INFO mapreduce.JobSubmitter: number of splits:2
24/04/03 05:16:57 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1712149730011_0001
24/04/03 05:16:58 INFO impl.YarnClientImpl: Submitted application application_1712149730011_0001
24/04/03 05:16:58 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1712149730011_0001/
24/04/03 05:16:58 INFO mapreduce.Job: Running job: job_1712149730011_0001
24/04/03 05:17:16 INFO mapreduce.Job: Job job_1712149730011_0001 running in uber mode : false
24/04/03 05:17:16 INFO mapreduce.Job: map 0% reduce 0%
24/04/03 05:17:59 INFO mapreduce.Job: map 67% reduce 0%
24/04/03 05:18:02 INFO mapreduce.Job: map 100% reduce 0%
24/04/03 05:18:21 INFO mapreduce.Job: map 100% reduce 100%
24/04/03 05:18:22 INFO mapreduce.Job: Job job_1712149730011_0001 completed successfully
24/04/03 05:18:22 INFO mapreduce.Job: Counters: 49
    File System Counters
        FILE: Number of bytes read=410
        FILE: Number of bytes written=356607
        FILE: Number of read operations=0
        FILE: Number of large read operations=0
        FILE: Number of write operations=0
        HDFS: Number of bytes read=560
        HDFS: Number of bytes written=281

```
        HDFS: Number of read operations=9
        HDFS: Number of large read operations=0
        HDFS: Number of write operations=2
    Job Counters
        Launched map tasks=2
        Launched reduce tasks=1
        Data-local map tasks=2
        Total time spent by all maps in occupied slots (ms)=91697
        Total time spent by all reduces in occupied slots (ms)=13191
        Total time spent by all map tasks (ms)=91697
        Total time spent by all reduce tasks (ms)=13191
        Total vcore-seconds taken by all map tasks=91697
        Total vcore-seconds taken by all reduce tasks=13191
        Total megabyte-seconds taken by all map tasks=93897728
        Total megabyte-seconds taken by all reduce tasks=13507584
    Map-Reduce Framework
        Map input records=1
        Map output records=40
        Map output bytes=324
        Map output materialized bytes=416
        Input split bytes=194
        Combine input records=0
        Combine output records=0
        Reduce input groups=34
        Reduce shuffle bytes=416
        Reduce input records=40
        Reduce output records=34
        Spilled Records=80
        Shuffled Maps =2
        Failed Shuffles=0
        Merged Map outputs=2
        GC time elapsed (ms)=1366
        CPU time spent (ms)=8040
        Physical memory (bytes) snapshot=653721600
        Virtual memory (bytes) snapshot=2367148032
        Total committed heap usage (bytes)=303374336
    Shuffle Errors
        BAD_ID=0
        CONNECTION=0
        IO_ERROR=0
        WRONG_LENGTH=0
        WRONG_MAP=0
        WRONG_REDUCE=0
    File Input Format Counters
        Bytes Read=366
    File Output Format Counters
        Bytes Written=281
24/04/03 05:18:22 INFO streaming.StreamJob: Output directory: /home/hari/result
```

rr@ubuntu:~$ hdfs dfs -ls /home/hari/

```
24/04/03 05:18:56 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
Found 9 items
-rw-r--r--   1 rr supergroup        244 2024-04-03 05:15 /home/hari/bigdata.txt
drwxr-xr-x   - rr supergroup          0 2024-04-03 05:18 /home/hari/result
-rw-r--r--   1 rr supergroup        148 2024-04-03 05:14 /home/hari/wordMapper.py
-rw-r--r--   1 rr supergroup        276 2024-04-03 05:15 /home/hari/wordReducer.py
```

rr@ubuntu:~$ hdfs dfs -ls /home/hari/result

```
24/04/03 05:19:19 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r--   1 rr supergroup          0 2024-04-03 05:18 /home/hari/result/_SUCCESS
-rw-r--r--   1 rr supergroup        281 2024-04-03 05:18 /home/hari/result/part-00000
```

rr@ubuntu:~$ hdfs dfs -cat /home/hari/result/part-00000

```
24/04/03 05:19:39 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
and      1
government,    1
industry,      1
is       1
some     2
advancements 1
analytics      2
in       2
its      1
Read     1
to       1
behind 1
other    1
health   1
more     1
used     1
Whether        1
big      2
of       1
finance,       1
most     1
significant    1
today.   1
data     2
on       1
about    1
benefits.      1
world    1
many     1
industry       1
```

```
learn	1
care,	1
the	2
or	1
```

rr@ubuntu:~$ stop-all.sh

```
This script is Deprecated. Instead use stop-dfs.sh and stop-yarn.sh
24/04/03 05:20:04 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
Stopping namenodes on [localhost]
localhost: stopping namenode
localhost: stopping datanode
Stopping secondary namenodes [0.0.0.0]
0.0.0.0: stopping secondarynamenode
24/04/03 05:20:36 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
stopping yarn daemons
stopping resourcemanager
localhost: stopping nodemanager
localhost: nodemanager did not stop gracefully after 5 seconds: killing with kill -9
no proxyserver to stop
```

SOURCE CODE:

rr@ubuntu:~$ start-all.sh
This script is Deprecated. Instead use start-dfs.sh and start-yarn.sh
24/04/03 08:30:11 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting namenodes on [localhost]
localhost: starting namenode, logging to /usr/local/hadoop/logs/hadoop-rr-namenode-ubuntu.out
localhost: starting datanode, logging to /usr/local/hadoop/logs/hadoop-rr-datanode-ubuntu.out
Starting secondary namenodes [0.0.0.0]
0.0.0.0: starting secondarynamenode, logging to /usr/local/hadoop/logs/hadoop-rr-secondarynamenode-ubuntu.out
24/04/03 08:30:54 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
starting yarn daemons
starting resourcemanager, logging to /usr/local/hadoop/logs/yarn-rr-resourcemanager-ubuntu.out
localhost: starting nodemanager, logging to /usr/local/hadoop/logs/yarn-rr-nodemanager-ubuntu.out

rr@ubuntu:~$ gedit emp_det.txt

```
1001,Hari,22,20000
1002,Bala,25,12000
1003,Shalini,23,10000
1004,Ravi,24,16000
1005,Vijay,23,13000
1006,Karthi,26,17000
```

rr@ubuntu:~$ hdfs dfsadmin -safemode leave
24/04/03 08:38:00 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Safe mode is OFF

rr@ubuntu:~$ hdfs dfs -put emp_det.txt /home/hari/
24/04/03 08:38:08 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

rr@ubuntu:~$ hdfs dfs -ls /home/hari/
24/04/03 08:39:01 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 item
-rw-r--r-- 1 rr supergroup 123 2024-04-03 08:38 /home/hari/emp_det.txt

rr@ubuntu:~$ pig

grunt> A = LOAD'/home/hari/emp_det.txt' USING PigStorage(',') as (id:int,name:chararray,age:int,salary:int);

grunt> DUMP A;

```
(1001,Hari,22,20000)
(1002,Bala,25,12000)
(1003,Shalini,23,10000)
(1004,Ravi,24,16000)
(1005,Vijay,23,13000)
(1006,Karthi,26,17000)
```

grunt> fewrecords = LIMIT A 3;

grunt> DUMP fewrecords;

```
(1001,Hari,22,20000)
(1002,Bala,25,12000)
(1003,Shalini,23,10000)
```

grunt> agelimit = FILTER A BY age>25;

grunt> DUMP agelimit;

```
(1006,Karthi,26,17000)
```

grunt> DESCRIBE A;

```
A: {id: int,name: chararray,age: int,salary: int}
```

grunt> groupbyage = GROUP A BY age;

grunt> DUMP groupbyage;

```
(22,{(1001,Hari,22,20000)})
(23,{(1005,Vijay,23,13000),(1003,Shalini,23,10000)})
(24,{(1004,Ravi,24,16000)})
(25,{(1002,Bala,25,12000)})
(26,{(1006,Karthi,26,17000)})
```

grunt> maxage = FOREACH groupbyage GENERATE(A.id,A.age),MAX(A.age);

grunt> DUMP maxage;

```
(({(1001)},{(22)}),22)
(({(1005),(1003)},{(23),(23)}),23)
(({(1004)},{(24)}),24)
(({(1002)},{(25)}),25)
(({(1006)},{(26)}),26)
```

grunt> allagevalue = GROUP A ALL;

grunt> DUMP allagevalue;

```
(all,{(1006,Karthi,26,17000),(1005,Vijay,23,13000),(1004,Ravi,24,16000),(1003,Shalini,23,1
0000),(1002,Bala,25,12000),(1001,Hari,22,20000)})
```

grunt> sumofsalary = FOREACH allagevalue GENERATE group,SUM(A.salary);

grunt> DUMP sumofsalary;
(all,88000)

grunt> dist = DISTINCT A;

grunt> DUMP dist;
(1001,Hari,22,20000)
(1002,Bala,25,12000)
(1003,Shalini,23,10000)
(1004,Ravi,24,16000)
(1005,Vijay,23,13000)
(1006,Karthi,26,17000)

grunt> quit;
2024-04-03 09:51:47,018 [main] INFO org.apache.pig.Main - Pig script completed in 1 hour, 11 minutes, 15 seconds and 274 milliseconds (4275274 ms)
rr@ubuntu:~$ stop-all.sh
This script is Deprecated. Instead use stop-dfs.sh and stop-yarn.sh
24/04/03 09:51:59 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Stopping namenodes on [localhost]
localhost: stopping namenode
localhost: stopping datanode
Stopping secondary namenodes [0.0.0.0]
0.0.0.0: stopping secondarynamenode
24/04/03 09:52:33 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
stopping yarn daemons
stopping resourcemanager
localhost: stopping nodemanager
localhost: nodemanager did not stop gracefully after 5 seconds: killing with kill -9
no proxyserver to stop

SOURCE CODE:

rr@ubuntu:~$ start-all.sh
This script is Deprecated. Instead use start-dfs.sh and start-yarn.sh
24/04/03 17:38:38 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting namenodes on [localhost]
localhost: starting namenode, logging to /usr/local/hadoop/logs/hadoop-rr-namenode-ubuntu.out
localhost: starting datanode, logging to /usr/local/hadoop/logs/hadoop-rr-datanode-ubuntu.out
Starting secondary namenodes [0.0.0.0]
0.0.0.0: starting secondarynamenode, logging to /usr/local/hadoop/logs/hadoop-rr-secondarynamenode-ubuntu.out
24/04/03 17:39:12 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
starting yarn daemons
starting resourcemanager, logging to /usr/local/hadoop/logs/yarn-rr-resourcemanager-ubuntu.out
localhost: starting nodemanager, logging to /usr/local/hadoop/logs/yarn-rr-nodemanager-ubuntu.out
rr@ubuntu:~$ gedit emp_det.txt

```
1001,Hari,22,20000
1002,Suriya,25,12000
1003,Arun,23,10000
1004,Jeeva,24,16000
1005,Karthi,23,13000
1006,Mohan,26,17000
```

rr@ubuntu:~$ gedit emp_det2.txt

```
1001,Kallakurichi,MCA,Developer
1002,ChinnaSalem,MBA,HR
1003,Trichy,MCOM,Sales
1004,Salem,MCOM,Ad
1005,Attur,MA,Marketing
1006,Karur,MSC,DE
```

rr@ubuntu:~$ gedit udf.py

```
@outputSchema('num:long')
def get_length(data):
    str_data=''.join([chr(x) for x in data])
    return len(str_data)
```

rr@ubuntu:~$ hdfs dfsadmin -safemode leave
24/04/03 17:42:56 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Safe mode is OFF
rr@ubuntu:~$ hdfs dfs -put emp_det1.txt /home/hari/
24/04/03 17:55:01 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

```
rr@ubuntu:~$ hdfs dfs -put emp_det1.txt /home/hari/
24/04/03 17:55:51 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable

rr@ubuntu:~$ pig

grunt> REGISTER 'udf.py' USING jython as pyudf

grunt> A = LOAD'/home/hari/emp_det.txt' USING PigStorage(',');

grunt> B = FOREACH A GENERATE $0, pyudf.get_length($0);

grunt> DUMP B;
        (1001,4)
        (1002,4)
        (1003,4)
        (1004,4)
        (1005,4)
        (1006,4)
grunt> quit

rr@ubuntu:~$ pig

grunt> A = LOAD'/home/hari/emp_det.txt' USING PigStorage(',') AS
(id:int,name:chararray,age:int,salary:int);

grunt> B = LOAD'/home/hari/emp_det2.txt' USING PigStorage(',') AS
(id:int,location:chararray,qualification:chararray,designation:chararray);

grunt> join_data = JOIN A by id,B by id;

grunt> SPLIT A INTO X IF age==22,Y IF age>=23;

grunt> DUMP X;
        (1001,Hari,22,20000)

grunt> DUMP Y;
        (1002,Suriya,25,12000)
        (1003,Arun,23,10000)
        (1004,Jeeva,24,16000)
        (1005,Karthi,23,13000)
        (1006,Mohan,26,17000)

grunt> grade_grp = GROUP A BY name;

grunt> grade_avg = FOREACH grade_grp GENERATE A.name,A.age,AVG(A.salary);

grunt> DUMP grade_avg;
        ({(Arun)},{(23)},10000.0)
        ({(Jeeva)},{(24)},16000.0)
```

```
(({(Mohan)},{(26)},17000.0)
(({(Karthi)},{(23)},13000.0)
(({(Suriya)},{(25)},12000.0)
(({(Hari)},{(22)},20000.0)
```

grunt> cogroup_data = COGROUP A BY id,B BY id;

grunt> DUMP cogroup_data;

```
(1001,{(1001,Hari,22,20000)},{(1001,Kallakurichi,MCA,Developer)})
(1002,{(1002,Suriya,25,12000)},{(1002,ChinnaSalem,MBA,HR)})
(1003,{(1003,Arun,23,10000)},{(1003,Trichy,MCOM,Sales)})
(1004,{(1004,Jeeva,24,16000)},{(1004,Salem,MCOM,Ad)})
(1005,{(1005,Karthi,23,13000)},{(1005,Attur,MA,Marketing)})
(1006,{(1006,Mohan,26,17000)},{(1006,Karur,MSC,DE)})
```

grunt> M = FOREACH A GENERATE TOMAP(name,age);

grunt> DUMP M;

```
([Hari#22])
([Suriya#25])
([Arun#23])
([Jeeva#24])
([Karthi#23])
([Mohan#26])
```

grunt> QUIT;

2024-04-03 18:34:50,244 [main] INFO org.apache.pig.Main - Pig script completed in 38 minutes, 26 seconds and 173 milliseconds (2306173 ms)

rr@ubuntu:~$ stop-all.sh

This script is Deprecated. Instead use stop-dfs.sh and stop-yarn.sh
24/04/03 18:35:02 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Stopping namenodes on [localhost]
localhost: stopping namenode
localhost: stopping datanode
Stopping secondary namenodes [0.0.0.0]
0.0.0.0: stopping secondarynamenode
24/04/03 18:35:32 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
stopping yarn daemons
stopping resourcemanager
localhost: stopping nodemanager
localhost: nodemanager did not stop gracefully after 5 seconds: killing with kill -9
no proxyserver to stop

SOURCE CODE:

rr@ubuntu:~$ start-all.sh
This script is Deprecated. Instead use start-dfs.sh and start-yarn.sh
24/04/03 08:03:18 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting namenodes on [localhost]
localhost: starting namenode, logging to /usr/local/hadoop/logs/hadoop-rr-namenode-ubuntu.out
localhost: starting datanode, logging to /usr/local/hadoop/logs/hadoop-rr-datanode-ubuntu.out
Starting secondary namenodes [0.0.0.0]
0.0.0.0: starting secondarynamenode, logging to /usr/local/hadoop/logs/hadoop-rr-secondarynamenode-ubuntu.out
24/04/03 08:04:10 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
starting yarn daemons
starting resourcemanager, logging to /usr/local/hadoop/logs/yarn-rr-resourcemanager-ubuntu.out
localhost: starting nodemanager, logging to /usr/local/hadoop/logs/yarn-rr-nodemanager-ubuntu.out

rr@ubuntu:~$ gedit wCount.pig
```
Input_data = LOAD '/home/hari/bigdata.txt' As (line:chararray);
token = Foreach Input_data generate flatten(TOKENIZE(line)) as word;
group_data = GROUP token by word;
count_word = foreach group_data generate group as word, COUNT(token) as count;
order_data = order count_word by count DESC;
DUMP order_data;
```

rr@ubuntu:~$ gedit bigdata.txt
Whether used in health care, government, finance, or some other industry, big data analytics is behind some of the most significant industry advancements in the world today. Read on to learn more about big data analytics and its many benefits.

rr@ubuntu:~$ hdfs dfsadmin -safemode leave
24/04/03 08:05:43 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Safe mode is OFF

rr@ubuntu:~$ hdfs dfs -put bigdata.txt /home/hari
24/04/03 08:07:08 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

rr@ubuntu:~$ pig wCount.pig
(in,2)
(analytics,2)
(industry,2)
(some,2)
(data,2)
(the,2)

(big,2)
(Whether,1)
(today.,1)
(health,1)
(behind,1)
(world,1)
(other,1)
(learn,1)
(about,1)
(used,1)
(on,1)
(most,1)
(more,1)
(many,1)
(of,1)
(care,1)
(Read,1)
(is,1)
(its,1)
(advancements,1)
(and,1)
(significant,1)
(government,1)
(benefits.,1)
(to,1)
(or,1)
(finance,1)

rr@ubuntu:~$ stop-all.sh
This script is Deprecated. Instead use stop-dfs.sh and stop-yarn.sh
24/04/03 08:24:37 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Stopping namenodes on [localhost]
localhost: stopping namenode
localhost: stopping datanode
Stopping secondary namenodes [0.0.0.0]
0.0.0.0: stopping secondarynamenode
24/04/03 08:25:15 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
stopping yarn daemons
stopping resourcemanager
localhost: stopping nodemanager
localhost: nodemanager did not stop gracefully after 5 seconds: killing with kill -9
no proxyserver to stop

SOURCE CODE:

rr@ubuntu:~$ start-all.sh
This script is Deprecated. Instead use start-dfs.sh and start-yarn.sh
24/04/03 07:11:17 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting namenodes on [localhost]
localhost: namenode running as process 6611. Stop it first.
localhost: datanode running as process 6745. Stop it first.
Starting secondary namenodes [0.0.0.0]
0.0.0.0: secondarynamenode running as process 6975. Stop it first.
24/04/03 07:11:34 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
starting yarn daemons
resourcemanager running as process 7113. Stop it first.
localhost: nodemanager running as process 7262. Stop it first.

rr@ubuntu:~$ hdfs dfsadmin -safemode leave
24/04/03 07:12:08 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Safe mode is OFF

rr@ubuntu:~$ gedit minMax.txt

```
2000,99
2000,70
2000,80
2001,22
2001,50
2002,34
2002,50
```

rr@ubuntu:~$ gedit minMax.pig

```
A = LOAD'/home/hari/minMax.txt' USING PigStorage(',') AS (year:int,temp:int);
B = GROUP A BY year;
C = FOREACH B GENERATE group,MAX(A.temp),MIN(A.temp);
DUMP C;
```

rr@ubuntu:~$ hdfs dfs -put minMax.txt /home/hari
24/04/03 07:13:30 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

rr@ubuntu:~$ pig minMax.pig

```
(2000,99,70)
(2001,50,22)
(2002,50,34)
```

rr@ubuntu:~$ stop-all.sh
This script is Deprecated. Instead use stop-dfs.sh and stop-yarn.sh
24/04/03 07:56:37 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Stopping namenodes on [localhost]
localhost: stopping namenode

```
localhost: stopping datanode
Stopping secondary namenodes [0.0.0.0]
0.0.0.0: stopping secondarynamenode
24/04/03 07:57:14 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
stopping yarn daemons
stopping resourcemanager
localhost: stopping nodemanager
localhost: nodemanager did not stop gracefully after 5 seconds: killing with kill -9
no proxyserver to stop
```

SOURCE CODE:

rr@ubuntu:~$ start-all.sh
This script is Deprecated. Instead use start-dfs.sh and start-yarn.sh
24/04/03 06:25:02 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting namenodes on [localhost]
localhost: starting namenode, logging to /usr/local/hadoop/logs/hadoop-rr-namenode-ubuntu.out
localhost: starting datanode, logging to /usr/local/hadoop/logs/hadoop-rr-datanode-ubuntu.out
Starting secondary namenodes [0.0.0.0]
0.0.0.0: starting secondarynamenode, logging to /usr/local/hadoop/logs/hadoop-rr-secondarynamenode-ubuntu.out
24/04/03 06:26:04 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
starting yarn daemons
starting resourcemanager, logging to /usr/local/hadoop/logs/yarn-rr-resourcemanager-ubuntu.out
localhost: starting nodemanager, logging to /usr/local/hadoop/logs/yarn-rr-nodemanager-ubuntu.out

rr@ubuntu:~$ hdfs dfsadmin -safemode leave
24/04/03 06:26:45 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Safe mode is OFF

rr@ubuntu:~$ hive
Logging initialized using configuration in jar:file:/usr/local/hive/lib/hive-common-1.2.2.jar!/hive-log4j.properties

hive> CREATE DATABASE IF NOT EXISTS STUDENTS COMMENT 'STUDENT Details' WITH DBPROPERTIES ('Creator'='HARI');
OK
Time taken: 2.216 seconds

hive> SHOW DATABASES;
OK
default
emp
students
Time taken: 0.633 seconds, Fetched: 3 row(s)

hive> DROP DATABASE EMP;
OK
Time taken: 0.674 seconds

hive> SHOW DATABASES;
OK
default
students

Time taken: 0.107 seconds, Fetched: 2 row(s)

hive> CREATE TABLE MCA_DET(ROLL_N0 INT,NAME STRING,MARK_1 INT,MARK_2 INT,TOTAL INT) ROW FORMAT DELIMITED FIELDS TERMINATED BY ',' STORED AS TEXTFILE;
OK
Time taken: 0.581 seconds

hive> LOAD DATA LOCAL INPATH '/home/rr/mcadet.txt' INTO TABLE MCA_DET;
Loading data to table default.mca_det
Table default.mca_det stats: [numFiles=1, totalSize=148]
OK
Time taken: 2.853 seconds

hive> SELECT * FROM MCA_DET;
OK

1001	ARUN	59	40	99
1002	BALA	49	30	79
1003	CHANDRAN	45	30	75
1004	DHEEEN	43	35	78
1005	RANJITH	39	33	72
1006	SURIYA	35	29	64
1007	HARI	60	40	100

Time taken: 0.693 seconds, Fetched: 7 row(s)

hive> SELECT*FROM MCA_DET WHERE TOTAL>75;
OK

1001	ARUN	59	40	99
1002	BALA	49	30	79
1004	DHEEEN	43	35	78
1007	HARI	60	40	100

Time taken: 0.466 seconds, Fetched: 4 row(s)

hive> CREATE TABLE MCA LIKE MCA_DET;
OK
Time taken: 0.219 seconds

hive> SELECT*FROM MCA;
OK
Time taken: 0.201 seconds

hive> INSERT INTO TABLE MCA SELECT*FROM MCA_DET;
Query ID = rr_20240403065430_2c276384-0a0b-4b9c-b2fb-dfe7990daa6a
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks is set to 0 since there's no reduce operator
Starting Job = job_1712154390757_0002, Tracking URL = http://localhost:8088/proxy/application_1712154390757_0002/
Kill Command = /usr/local/hadoop/bin/hadoop job -kill job_1712154390757_0002
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 0

```
2024-04-03 06:54:45,784 Stage-1 map = 0%,  reduce = 0%
2024-04-03 06:54:59,734 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 1.78 sec
MapReduce Total cumulative CPU time: 1 seconds 780 msec
Ended Job = job_1712154390757_0002
Stage-4 is selected by condition resolver.
Stage-3 is filtered out by condition resolver.
Stage-5 is filtered out by condition resolver.
Moving data to: hdfs://localhost:54310/user/hive/warehouse/mca/.hive-staging_hive_2024-
04-03_06-54-30_693_4096659787887499699-1/-ext-10000
Loading data to table default.mca
Table default.mca stats: [numFiles=1, numRows=7, totalSize=148, rawDataSize=141]
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1   Cumulative CPU: 1.78 sec   HDFS Read: 3859 HDFS Write: 216
SUCCESS
Total MapReduce CPU Time Spent: 1 seconds 780 msec
OK
Time taken: 31.828 seconds
```

hive> SELECT*FROM MCA;

```
OK
1001    ARUNv       59      40      99
1002    BALA        49      30      79
1003    CHANDRAN    45      30      75
1004    DHEEEN      43      35      78
1005    RANJITH     39      33      72
1006    SURIYA      35      29      64
1007    Hari        60      40      100
Time taken: 0.174 seconds, Fetched: 7 row(s)
```

hive> QUIT;

rr@ubuntu:~$ stop-all.sh

```
This script is Deprecated. Instead use stop-dfs.sh and stop-yarn.sh
24/04/03 06:55:39 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
Stopping namenodes on [localhost]
localhost: stopping namenode
localhost: stopping datanode
Stopping secondary namenodes [0.0.0.0]
0.0.0.0: stopping secondarynamenode
24/04/03 06:56:14 WARN util.NativeCodeLoader: Unable to load native-hadoop library for
your platform... using builtin-java classes where applicable
stopping yarn daemons
stopping resourcemanager
localhost: stopping nodemanager
localhost: nodemanager did not stop gracefully after 5 seconds: killing with kill -9
no proxyserver to stop
```

SOURCE CODE:

```
test> use student;

student> show dbs;

student> db.createCollection('mca');

student> db.mca.insertOne({'_id':1,'Name':'Hari ','Roll No':'MCA001','Mark 1':78,'Mark
2':89,'Total':167});


student> db.mca.insertMany([{'_id':2,'Name':'Muthees','Roll No':'MCA002','Mark
1':67,'Mark 2':78,'Total':145},{'_id':3,'Name':'Tamil','Roll No':'MCA003','Mark 1':70,'Mark
2':88,'Total':158},{'_id':4,'Name':'Saran','Roll No':'MCA004','Mark 1':80,'Mark
2':60,'Total':140},{'_id':5,'Name':'Sabari','Roll No':'MCA005','Mark 1':70,'Mark
2':77,'Total':147},{'_id':6,'Name':'Selvi','Roll No':'MCA006','Mark 1':80,'Mark
2':50,'Total':130},{'_id':7,'Name':'Gayu','Roll No':'MCA008','Mark 1':90,'Mark
2':70,'Total':160}]);


student> db.mca.updateOne({'_id':7},{$set:{'Roll No':'MCA007'}});

student> db.mca.updateMany({},{$set:{'Batch':2024}});

student> db.mca.deleteOne({'_id':6});

student> db.mca.find();

student> db.dropDatabase();
```

OUTPUT:

- Access Database:

```
switched to db student
```

- List Database:

```
Sample   72.00 KiB
admin    72.00 KiB
config   72.00 KiB
local    88.00 KiB
```

- Create Collection:

```
{ ok: 1 }
```

- Insert One Document:

```
{
  acknowledged: true,
  insertedId: 1
}
```

- Insert Many Documents:

```
{
  acknowledged: true,
  insertedIds: {
    '0': 2,
    '1': 3,
    '2': 4,
    '3': 5,
    '4': 6,
    '5': 7
  }
}
```

- Update One Document:

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

- Insert Many Documents:

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 7,
  modifiedCount: 7,
  upsertedCount: 0
}
```

- Delete One Document:

```
{
  acknowledged: true,
  deletedCount: 1
}
```

- Find Documents:

```
{
  _id: 1,
  Name: 'Hari ',
  'Roll No': 'MCA001',
  'Mark 1': 78,
  'Mark 2': 89,
  Total: 167,
  Batch: 2024
}
{
  _id: 2,
  Name: 'Muthees',
  'Roll No': 'MCA002',
  'Mark 1': 67,
  'Mark 2': 78,
  Total: 145,
  Batch: 2024
}
{
  _id: 3,
  Name: 'Tamil',
  'Roll No': 'MCA003',
  'Mark 1': 70,
  'Mark 2': 88,
  Total: 158,
  Batch: 2024
}
{
  _id: 4,
  Name: 'Saran',
  'Roll No': 'MCA004',
  'Mark 1': 80,
  'Mark 2': 60,
  Total: 140,
  Batch: 2024
}
```

```
{
  _id: 5,
  Name: 'Sabari',
  'Roll No': 'MCA005',
  'Mark 1': 70,
  'Mark 2': 77,
  Total: 147,
  Batch: 2024
}
{
  _id: 7,
  Name: 'Gayu',
  'Roll No': 'MCA007',
  'Mark 1': 90,
  'Mark 2': 70,
  Total: 160,
  Batch: 2024
}
```

- Drop Database:

```
{ ok: 1, dropped: 'student' }
```

SOURCE CODE:

```
test> use student;
student> db.mca.find({Name:'Hari '});
student> db.mca.find({$or:[{Name:'Muthees'},{Total:64}]});
student> db.mca.find({Name:{$in:['Tamil','Hari ']}});
student> db.mca.find({Total:{$gt:165}});
student> db.mca.find({Total:{$lt:140}});
student> db.mca.find({Total:{$eq:160}});
student> db.mca.find({Name:{$nin:['Saran','Sabari','Selvi','Hari ', 'Gayu']}});
```

OUTPUT:

- Find Specific Document:

```
{
  _id: 1,
  Name: 'Hari ',
  'Roll No': 'MCA001',
  'Mark 1': 78,
  'Mark 2': 89,
  Total: 167,
  Batch: 2024
}
```

- OR Operator:

```
{
  _id: 2,
  Name: 'Muthees',
  'Roll No': 'MCA002',
  'Mark 1': 67,
  'Mark 2': 78,
  Total: 145,
  Batch: 2024
}
```

- IN Operator:

```
{
  _id: 3,
  Name: 'Tamil',
  'Roll No': 'MCA003',
  'Mark 1': 70,
  'Mark 2': 88,
  Total: 158,
  Batch: 2024
}
{
  _id: 1,
  Name: 'Hari ',
  'Roll No': 'MCA001',
  'Mark 1': 78,
  'Mark 2': 89,
  Total: 167,
  Batch: 2024
}
```

- Greater than Operator:

```
{
  _id: 1,
  Name: 'Hari ',
  'Roll No': 'MCA001',
  'Mark 1': 78,
  'Mark 2': 89,
  Total: 167,
  Batch: 2024
}
```

- Less than Operator:

```
{
  _id: 6,
  Name: 'Selvi',
  'Roll No': 'MCA006',
  'Mark 1': 80,
  'Mark 2': 50,
  Total: 130,
  Batch: 2024
}
```

- Equal to Operator:

```
{
  _id: 7,
  Name: 'Gayu',
  'Roll No': 'MCA007',
  'Mark 1': 90,
  'Mark 2': 70,
  Total: 160,
  Batch: 2024
}
```

- NIN Operator:

```
{
  _id: 2,
  Name: 'Muthees',
  'Roll No': 'MCA002',
  'Mark 1': 67,
  'Mark 2': 78,
  Total: 145,
  Batch: 2024
}
{
  _id: 3,
  Name: 'Tamil',
  'Roll No': 'MCA003',
  'Mark 1': 70,
  'Mark 2': 88,
  Total: 158,
  Batch: 2024
}
```